

Lesson 3: Ohm's Law

ChatGPT edition — prove the circuit

Meter discipline, circuit evidence, and acceptance proof

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- Switched holder for 2 AA cells
- 100, 220, and 470 Ω resistors
- Digital multimeter with fused current input
- Four insulated clip leads
- Graph paper and pencil
- Optional second meter for simultaneous readings

Predict first

For the same source, rank the three resistors from greatest to least current. Then predict whether V/I will agree with the marked resistance within the combined tolerances.

METER STATE IS PART OF THE CIRCUIT

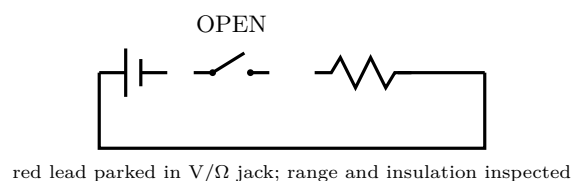
An ammeter is a low-resistance path. Never place it directly across a cell or battery holder. Move the red lead back to the voltage/resistance jack immediately after every current measurement. Change connections only with the holder switched off and disconnected. If a meter, lead, cell, or resistor warms, disconnect at once; do not repeat until the cause is found.

Questions before touching a lead

1. Which meter jack is fused?
2. What is the fuse rating printed on the meter?
3. Does the selected current range exceed the predicted current?
4. Which measurements require an energized circuit?
5. Which require an isolated, unpowered component?
6. Why does resistance mode supply its own small test current?
7. Where must a voltmeter connect: in series or in parallel?
8. Where must an ammeter connect?
9. What reading would “OL” mean on each selected function?
10. What single action ends every abnormal test?

Six connection states

State 0 — inspect, de-energized



Cells installed, switch open. Confirm resistor identity, sound insulation, firm clips, correct jacks, and a current estimate below the meter fuse and selected range.

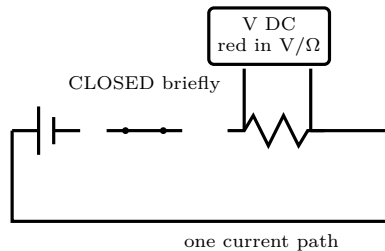
State 1 — resistance, component isolated

one resistor, disconnected



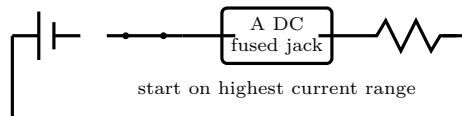
Resistance mode only on an isolated component. Touch the probes together first and record lead/contact resistance. Never use Ω or continuity mode on an energized circuit.

State 2 — voltage, meter in parallel



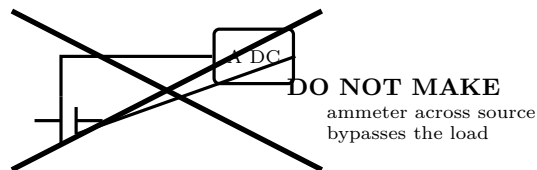
The voltmeter compares the two resistor terminals. Start on a range above the source voltage. Open the switch immediately after the reading.

State 3 — current, meter inserted in series



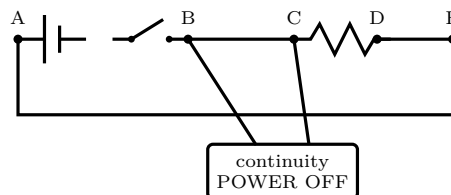
Open the original path and insert the meter so all current passes through it. Power only long enough for a stable reading. Switch off, disconnect, then park the red lead back in the V/ Ω jack.

State 4 — forbidden short-circuit state



Do not “see what happens.” This state can blow the meter fuse, heat leads or cells, and erase the evidence needed to diagnose the mistake.

State 5 — fault isolation, power off first



With cells removed, test one segment at a time. A continuity tone answers “low enough for this meter,” not “zero ohms” and not “safe under load.”

Prediction sheet

Marked R	tolerance	predicted $I = V_s/R$	predicted V_R	predicted $P = V_s^2/R$	What could make the result differ?
100 Ω	_____	_____	_____	_____	_____
220 Ω	_____	_____	_____	_____	_____
470 Ω	_____	_____	_____	_____	_____

Controlled verification

1. Measure the open-circuit source voltage V_s . Record meter function, range, jacks, resolution, and reading.
2. For each resistor, perform State 1 and record R_{meter} .
3. Build State 2. Close briefly; record loaded V_R . Open and disconnect.
4. Reconfigure to State 3. Close briefly; record I . Open and disconnect.
5. Repeat each energized reading once after rebuilding the connection. Do not average away a disagreement; investigate it.
6. Calculate $R_{\text{calc}} = V_R/I$ and $P = V_R I$.
7. Plot V_R vertically against I horizontally. A fitted slope is an estimate of resistance; do not force the line through zero without justification.

Evidence table

Marked R	R_{meter}	V_R run 1/2	I run 1/2	R_{calc}	P	warm?	Meter function, range, jacks, resolution
100 Ω							
220 Ω							
470 Ω							

Source behavior			
Open-circuit V_s	Loaded V_s , 100 Ω	Loaded V_s , 220 Ω	Loaded V_s , 470 Ω
If loaded source voltage changes with current, explain why treating V_s as one exact constant would distort the prediction.			

Worked calculation

Suppose $V_R = 2.91$ V and $I = 13.1$ mA = 0.0131 A:

$$R_{\text{calc}} = \frac{V_R}{I} = \frac{2.91 \text{ V}}{0.0131 \text{ A}} = 222 \Omega, \quad P = V_R I = (2.91)(0.0131) = 0.0381 \text{ W}.$$

For a 220 Ω , 5% resistor, the marked interval is $220(1 \pm 0.05) = 209$ to 231 Ω . The calculated value lies inside it, but that agreement is not yet proof: meter uncertainty and repeatability also belong in the comparison.

Uncertainty without false precision

Use the accuracy statement from the meter manual. If it says $\pm(a\% \text{ of reading} + b \text{ digits})$, then

$$u_x = (a/100)|x| + b(\text{least significant digit}).$$

Example: on a display reading 2.91 V, with accuracy $\pm(0.5\% + 2 \text{ digits})$ and resolution 0.01 V,

$$u_V = 0.005(2.91) + 2(0.01) = 0.0346 \text{ V}, \quad V = 2.91 \pm 0.035 \text{ V}.$$

A conservative bound for division is

$$\frac{u_R}{R} \leq \frac{u_V}{|V|} + \frac{u_I}{|I|}.$$

Record the bound as an interval. Round the reported result to the same decimal place as its uncertainty.

Uncertainty record					
Part	u_V	u_I	$R_{\text{calc}} \pm u_R$	marked interval	Do the intervals overlap?
100 Ω					
220 Ω					
470 Ω					

Fault isolation: preserve the evidence

Observation	Power-off checks first	Then, if checks pass
No current; no voltage across resistor	Open switch, loose clip, depleted/reversed cell, broken lead	Measure source voltage; trace voltage from A to E
Voltage across resistor; zero current indication	Wrong function/range/jack, blown current fuse, open meter lead	Verify meter on a known bounded test path
Full source voltage across one connection	Open circuit immediately downstream	Power off; continuity-test that segment only
Near-zero resistor voltage; unexpected current	Wrong resistor, bypass lead, solder/clip bridge	Remove cells; inspect and measure isolated resistance
Reading changes when leads move	Loose/oxidized contact or damaged lead	Replace one lead, rebuild, repeat both runs
Resistor, cells, lead, or meter warms	Stop; disconnect; do not retest warm hardware	Adult finds overcurrent or wrong connection before reuse

Challenge questions

- Why is V/I a stronger test than current alone?
- Why measure loaded source voltage?
- What does the slope of the V -versus- I graph mean?
- Which axis assignment makes the slope equal R ?
- Does a straight line prove the resistor is ohmic at every temperature?
- How can lead resistance matter most for the smallest resistor?
- Why might resistance mode and V/I disagree slightly?
- How does ammeter burden voltage alter the circuit?
- Why can a continuity tone coexist with a bad loaded connection?
- When is “OL” evidence of an open path, and when is it only over-range?
- Which digits in your calculation are not supported by the meters?
- What observation would make you reject rather than average a run?
- Does overlap of intervals prove the marked value is correct?
- What additional independent check would strengthen the claim?

Acceptance proof

ACCEPT ONLY IF EVERY CLAIM HAS EVIDENCE		
Criterion	pass?	Evidence location or value
All work stayed at isolated 2-AA power	☐	
Meter jacks/functions/ranges recorded	☐	
No heating, odor, damage, or blown fuse	☐	
Two runs agree within stated resolution	☐	
Current ranking matches prediction or is explained	☐	
R_{calc} and marked intervals compared	☐	
Graph axes, units, points, and slope are present	☐	
One fault scenario was reasoned through power-off	☐	
Red lead is parked in V/Ω jack	☐	
Cells are removed for storage	☐	

Claim: Within the stated conditions and uncertainty, the evidence ☐ supports ☐ does not support $V = IR$ for these resistors.

Learner initials _____ Adult inspection _____ Date _____

Controlled extension

Add a second resistor in series, then in parallel. Before connecting, predict equivalent resistance and maximum current. Accept the new circuit only after repeating States 0–3, the uncertainty comparison, and the acceptance proof. An LED is not the acceptance standard for Ohm's law: it is nonlinear and polarity-sensitive.

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.

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